

"Goudsmit and myself hit upon this idea by studying a paper of Pauli, in which the famous exclusion principle (to be treated in Chapter 9) was formulated and in which, for the first time, *four* quantum numbers were ascribed to the electron. This was done rather formally; no concrete picture was connected with it. To us this was a mystery. We were so conversant with the proposition that every quantum number corresponds to a degree of freedom (an independent coordinate), and on the other hand with the idea of a point electron, which obviously had three degrees of freedom only, that we could not place the fourth quantum number. We could understand it only if the electron was assumed to be a small sphere that could rotate. . . .

Somewhat later we found in a paper of Abraham, to which Ehrenfest drew our attention, that for a rotating sphere with surface charge the necessary factor two in the magnetic moment ($g_s = 2$) could be understood classically. This encouraged us, but our enthusiasm was considerably reduced when we saw that the rotational velocity at the surface of the electron had to be many times the velocity of light! I remember that most of these thoughts came to us on an afternoon at the end of September 1925. We were excited, but we had not the slightest intention of publishing anything. It seemed so speculative and bold, that something ought to be wrong with it, especially since Bohr, Heisenberg, and Pauli, our great authorities, had never proposed anything of the kind. But of course we told Ehrenfest. He was impressed at once, mainly, I feel, because of the visual character of our hypothesis, which was very much

in his line. He called our attention to several points, e.g., to the fact that in 1921 A. H. Compton already had suggested the idea of a spinning electron as a possible explanation of the natural unit of magnetism, and finally said that it was either highly important or nonsense, and that we should write a short note for *Naturwissenschaften* (a physics research journal) and give it to him. He ended with the words 'and then we will ask Lorentz.' This was done. Lorentz received us with his well known great kindness, and he was very much interested, although, I feel, somewhat skeptical too. He promised to think it over. And in fact, already next week he gave us a manuscript, written in his beautiful handwriting, containing long calculations on the electromagnetic properties of rotating electrons. We could not fully understand it, but it was quite clear that the picture of the rotating electron, if taken seriously, would give rise to serious difficulties. For one thing, the magnetic energy would be so large that by the equivalence of mass and energy the electron would have a larger mass than the proton, or, if one sticks to the known mass, the electron would be bigger than the whole atom! In any case, it seemed to be nonsense. Goudsmit and myself both felt that it might be better for the present not to publish anything; but when we said this to Ehrenfest, he answered: 'I have already sent your letter in long ago; you are both young enough to allow yourselves some foolishness!' " (from *The Conceptual Development of Quantum Mechanics* by Max Jammer, McGraw-Hill, 1966)